

Tortuosity in Porous Media: A Critical Review

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The concept of tortuosity is used to characterize the structure of porous media, to estimate their electrical and hydraulic conductivity, and to study the travel time and length for tracer dispersion, but different types of tortuosity—geometric, hydraulic, electrical, and diffusive—have been used essentially interchangeably in the literature. Here, we critically review the tortuosity models developed empirically, analytically, and numerically for flow in both saturated and unsaturated porous media. We emphasize that the proposed tortuosity models are distinct and thus may not be used interchangeably. Given the variety of models that have been developed, and the sharp differences between some of them, no consensus has emerged unifying the models in a coherent way. Related treatments of tortuosity are found in the literature on porous catalysts. In such materials, nonlinear reactions ordinarily accompany transport, and the effective diffusivity within the pore space in the presence of the reactions is distinct from the one in their absence. Thus, because tortuosity may be defined as the ratio of the effective diffusivities in the bulk material and within the pore space, a careful treatment of tortuosity may need to distinguish between transport with and without reactions. This complication is ultimately relevant to soils as well, because bioremediation and biodegradation in soils are always accompanied by nonlinear reactions. Common models of tortuosity include both logarithmic functions and power laws. In many cases, the differences between the logarithmic and power-law phenomenologies are not great, but power laws can usually be reconciled with percolation concepts. Invoking percolation theory provides both insight into the origin of the power functions and a framework for understanding differences between tortuosity models.

Porous media, particularly natural ones such as rock and soil, constitute a broad class of complex systems. Their pore space is highly chaotic, with pore sizes that vary across a broad range. But what makes porous media particularly complex is that the paths that they provide for fluid flow are not straight, but tortuous and meandering. A molecule often must traverse a path that is several times longer than the straight line between its original source and its destination. Not surprisingly, until the early 1970s pore-scale modeling of porous media and the flow and transport processes that take place in them was considered a hopeless task, precisely due to the complex morphology of the pore space. Fluid flow and transport processes were often modeled by averaging the microscopic conservation laws—the continuity and momentum equations—over a suitably selected segment of the pore space, such that on the scale of that segment a porous medium could be considered homogeneous. Such equations then contained effective flow and transport coefficients, such as the effective permeability, diffusivity, and electrical and thermal conductivities. These coefficients had to be measured experimentally for each medium—unless they could be predicted based on some hypothetical model of the pore space. Due to the complexity of pore space morphology, the most used model was the bundle of capillary tubes, in which the tubes, representing the pores,

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were arranged either in series or parallel (Scheidegger, 1974). While such models were amenable to the analytical derivation of expressions for the effective flow and transport properties, the predictions that they provided rarely agreed with experimental data, simply because the structure (e.g., interconnectivity) of the pore space of almost any natural porous medium was nothing like a bundle of tubes.

To address this glaring shortcoming, the concept of tortuosity was introduced by Carman (1937). The purpose (Dullien, 1979) was to match the permeability, calculated based on the bundle of capillary tubes, to experimental data. In effect, all relevant length scales were divided by the tortuosity to make them longer than what was provided by the bundle of capillary tubes so as to approximate the actual length of the paths that fluid molecules travel through the pore space.

While the concept of tortuosity seems straightforward, in practice tortuosity is not consistently defined, and its treatment in the literature is often misleading (Tye, 1983; Epstein, 1989; Sahimi, 1993; Moldrup et al., 2001). It is sometimes claimed that tortuosity depends on the type of flow or transport process being studied, and comparisons between predicted and measured fluid flow and transport are used to support that assertion (Moldrup et al., 2001; Vervoot and Cattle, 2003). Such a claim for the nature of tortuosity suggests, however, that its use in practice is often as an adjustable parameter—a “fudge factor”—hence implying that a clear understanding of tortuosity is lacking.

A fundamental question, rarely addressed, is whether tortuosity is an intrinsic property of the medium, of a process within the medium, or neither, being simply an ad hoc parameter used to improve the agreement between theory and experiment. Because the tortuosity has a pronounced saturation dependence, tortuosity cannot simply be a property of the medium itself but must be derived from the actual paths of flow, conduction, or transport involved. This review provides a comprehensive discussion of the concept of tortuosity, its varying definitions in the literature, and their significance. It also provides a conceptual background to help the reader organize the various results discussed for tortuosity. The review was motivated by the observation that the many definitions of tortuosity do not fulfill their intended purpose in the calculation and prediction of flow and transport properties of porous media and may in fact contribute to further confusion. As already pointed out, there are several types of tortuosity in the literature; we begin by discussing them. Our discussion necessarily omits many details of the experimental and numerical procedures, but references are provided for those wanting more information.

TYPES OF TORTUOSITY

In the literature, tortuosity has been defined as either a geometric parameter or one that is related to the hydraulic, electrical, or diffusive properties (Scheidegger, 1974; Sahimi, 1993; Clennell, 1997; Matyka et al., 2008). Geometric tortuosity, τ_g , is the ratio of the average length, $\langle L_g \rangle$, of the geometric flow

paths through the medium to the straight-line length, L_s , across the medium ($\tau_g > 1$):

$$\tau_g = \frac{\langle L_g \rangle}{L_s} \quad [1]$$

Tortuosity could instead be defined as the ratio of the shortest pathway ($L_{\min}$) to the straight-line length L_s (Adler, 1992); however, the form presented in Eq. [1] is the more general definition. Note that $L_{\min}$, the shortest pathway, is different from the chemical length, the length of the path corresponding to the minimal travel time of tracer particles that are dispersed in flow through a pore space (Lee et al., 1999). The geometric tortuosity coefficient may also be expressed as the inverse of the above definition ($T_g = 1/\tau_g$), and it is always the case that $T_g < 1$ (Hillel, 2004).

There are two main alternatives for calculating the average length of the flow paths. One may average across the actual lengths of all the flow lines themselves, disregarding the fact that fluid particles move along these flow lines at different velocities. Alternatively, one may average the lengths of the flow lines of all fluid particles passing through a given cross-section during a given period of time, giving a flux-weighted average. The latter alternative appears more natural when fluid flow in porous media is considered (Koponen et al., 1996). Other approaches have also been used, such as estimating the average path length from the pore size distribution (e.g., Feng and Yu, 2007).

The hydraulic tortuosity, τ_h , is the square of the ratio of the flux-weighted average path length for hydraulic flow, $\langle L_h \rangle$, to the straight-line length (Clennell, 1997). Bear (1972) defined the (hydraulic) tortuosity coefficient as the square of the inverse of that ratio, $T_h = (L_s/\langle L_h \rangle)^2$, with reported literature values in the range of 0.56 to 0.8 for the ratio itself (Bear, 1972). In the definition of tortuosity given by Carman (1937, 1956), which was followed by Bear (1972) and Epstein (1989) among others, tortuosity is not simply a ratio of distances but also involves projecting the local potential gradient along a flow path onto a vector parallel to the macroscopic gradient; this is the origin of the square of the length ratio used by Bear (1972) and others. In other words, based on the Dupuit relation, for isotropically and randomly distributed pore space we might expect that the pore velocity parallel to the direction of flow is u/ϕ (where u is the Darcy flux and ϕ is the porosity), where the fractional free area is ϕ . If a random pore space does not exist, the Dupuit assumption should be modified. Therefore, the pore velocity must instead be taken as $(u/\phi)(\langle L_h \rangle/L_s)$ because the actual flow path is tortuous (Carman, 1956). Carman (1956, p. 12) postulated that the time taken for a fluid particle to cross a tortuous pathway at a velocity $(u/\phi)(\langle L_h \rangle/L_s)$ corresponds to the time taken to cross a distance L_s at a velocity u/ϕ . This tortuosity coefficient has often been used as an additional input, beyond the effects of the pore-size distribution, for predicting the hydraulic conductivity (Dullien, 1979).

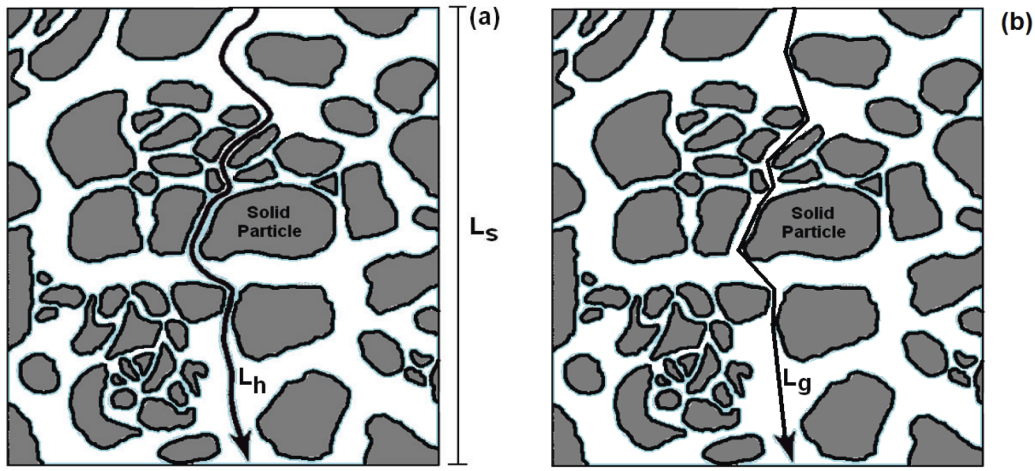


Fig. 1. Schematic of a possible (a) hydraulic (length L_h) and (b) geometric (length L_g) flow pathway in a saturated porous medium (the porous medium scheme is from Gvirtzman and Gorelick, 1991). Note that L_g is shorter than L_h because L_g takes shortcuts that cross streamlines. As a consequence, the geometric tortuosity $\tau_g < \tau_h$.

When experimental results do not conform to predictions from Carman (1937) based on the aforementioned bundle of tubes model, the discrepancy is frequently interpreted in terms of tortuosity. But inferring the hydraulic tortuosity τ_h by comparing the Kozeny–Carman (Kozeny, 1927; Carman, 1937) prediction with experiment is not supported by experimental data for all porous media: the Kozeny–Carman model does not hold for complex porous materials (Wyllie and Gregory, 1955; Carman, 1956; Bernabé and Bruderer, 1998). For media whose solid particles deviate strongly from a spherical shape and/or whose particle size distributions are broad, and for consolidated materials, the Kozeny–Carman equation is not valid (Carman, 1956, p. 13; Dullien, 1979, p. 171). In addition, the Kozeny–Carman equation based on the assumption of conduit (pipe or tube) flow is not valid for porous media with high porosities (e.g., $\phi \geq 0.95$), where the frictional drag dominates the viscous shear and, consequently, the resistance to flow (Dullien, 1979, p. 172). Thus, comparison of experimental data with the Kozeny–Carman prediction does not so much yield a tortuosity value (in the sense of a length ratio squared) as a rationale for invoking it as an adjustable parameter.

In general, geometric tortuosity and hydraulic tortuosity are not the same, and $\tau_g < \tau_h$. Imagine running a thread through a porous medium such that it lies precisely along a streamline (Fig. 1a). That is one possible hydraulic flow pathway L_h . If one then pulls the thread tight, without moving any solid particles, the result is the corresponding geometric flow pathway, L_g (Fig. 1b), which is shorter because it takes shortcuts that cross streamlines: hydraulic flow paths are smooth curves rather than straight lines or close tangents to the solid particles (Clennell, 1997).

Consider the direct-current (DC) electrical conductivity. The electrical tortuosity, τ_e , in the spirit of Carman's definition, can be defined as (Childs, 1969; Tindall et al., 1999; Wong, 1999)

$$\tau_e = \left(\frac{\langle L_e \rangle}{L_s} \right)^2 \quad [2]$$

where $\langle L_e \rangle$ is the average path length for electrical flow.

The electrical tortuosity can be also inferred by measuring the electrical resistivity of a medium as the product of its porosity ϕ and the formation factor F (Wyllie, 1957; Schopper, 1966; Dullien, 1979; Coleman and Vassilicos, 2008):

$$\tau_e = \phi F \quad [3]$$

where F is the quotient of the electrical resistivity of the saturated porous medium ρ_p and the resistivity of the saturating liquid ρ_l . The formation factor is a dimensionless quantity whose value is always > 1 in the absence of solid and/or surface conduction. In such cases, its value is supposedly determined uniquely by the pore geometry (Dullien, 1979), although pore topology (connectivity) must also be relevant. A form of Eq. [3] with an empirical exponent η has also been reported: $[\tau_e]_{\text{sat}} = (\phi F)^\eta$ (Shen and Chen, 2007). For example, Winsauer et al. (1952) measured the transit time of ions flowing through saturated porous media under a sufficiently high electric potential gradient (to increase the rate of ionic migration relative to the speed of diffusing molecules), calculated the ratio $\langle L_e \rangle / L_s$, and experimentally found that $\phi F = (\langle L_e \rangle / L_s)^{1.67}$, which is not greatly different from $\phi F = (\langle L_e \rangle / L_s)^2$, the combination of Eq. [2] and [3].

Equation [3] has been used to estimate hydraulic tortuosity for the Kozeny–Carman equation. Wyllie and Rose (1950) used $\phi F = (\langle L_h \rangle / L_s)$, while Cornell and Katz (1953) used $\phi F = (\langle L_h \rangle / L_s)^2$. In the study of Wyllie and Rose (1950), the equivalent Kozeny–Carman pipe had a constant cross-sectional area ϕA (where A is the sample cross-sectional area), with an effective length L_{eff} whereas Cornell and Katz (1953) considered a pipe with constant cross-sectional area $\phi A L / L_{\text{eff}}$ and the same effective length as Wyllie and Rose (1950) (noted by Wyllie and Gregory, 1955).

Lorenz (1961) demonstrated that the geometric tortuosity τ_g of flow lines accounting for both sinuosity and converging–diverging pore geometry was smaller than the electrical tortuosity: $\tau_g < \tau_e = \phi F$. Note that, in contrast to the Kozeny–Carman

formulation for the hydraulic conductivity, Eq. [3] does not explicitly account for a pore size distribution influence on the electrical conductivity. This distinction is based on the argument that the electrical conductivity of a given pore is independent of its radius, a justification that has, however, been shown to be appropriate only in relatively homogeneous media (Hunt and Ewing, 2009).

Analogously to electrical tortuosity, diffusive tortuosity, τ_d , can be defined (Epstein, 1989) as

$$\tau_d = \left(\frac{\langle L_d \rangle}{L_s} \right)^2 \quad [4]$$

where $\langle L_d \rangle$ is the average length of a chemical's diffusive pathway.

The diffusive tortuosity τ_d is sometimes defined as the diffusion coefficient of diffusing species in free fluid, d_f , relative to its value in a porous medium, d_p (Satterfield and Sherwood, 1963; Greenkorn, 1983; Clennell, 1997; Shen and Chen, 2007; Sahimi, 1993, 2011):

$$\tau_d = \frac{d_f}{d_p} \quad [5]$$

which results in a tortuosity term that includes a direct porosity effect. An alternative relation, common in soil science and analogous to Eq. [3], removes porosity from the tortuosity value in Eq. [5] (Currie, 1960; Epstein, 1989; Moldrup et al., 2001; Hillel, 2004; Kirkham, 2005):

$$\tau_d = \phi \frac{d_f}{d_p} \quad [6]$$

Equation [6] is the theoretical result of Currie (1970) (cited in Troeh et al., 1982), who applied Fick's first law to describe diffusion in a tortuous channel. Moldrup et al. (2001) substituted θ for ϕ to apply Eq. [6] to unsaturated conditions. The resulting parameter $1/\tau_d$ is sometimes called *pore continuity* in the gas diffusion literature (Moldrup et al., 2001).

In contrast to Eq. [5], Eq. [6] accounts directly for the porosity (or water content in unsaturated media) contribution, in-

terpreting d_p as depending on both the tortuosity of the medium and on the fractional volume (water content) or porosity (Hillel, 2004). Tortuosity, however, is itself a function of porosity or water content. Equations [5] and [6] also illustrate how tortuosity has been defined and applied inconsistently in the literature.

The diffusive tortuosity, τ_d , including the effects of chemical interactions, must depend on the molecular size of the diffusing particles (Sahimi, 2011) because some paths may become inaccessible to larger ones. If the molecular size becomes comparable to the pore size, the term d_f/d_p is multiplied by a dimensionless constrictivity factor δ (Boving and Grathwohl, 2001). In a different interpretation of constrictivity, van Brakel and Heertjes (1974) used a constrictivity factor to correct for the cross-section of a pore varying over its length or, more specifically, for transport being restricted by the pore necks. Their δ is largely a function of the ratio of the pore's maximum and minimum cross-sections (van Brakel and Heertjes, 1974).

For special conditions, e.g., steady-state diffusion under an established concentration gradient and volume or time-averaged flux in one-dimensional transport, Clennell (1997) demonstrated that diffusive and electrical tortuosity are identical, implying that $\langle L_e \rangle = \langle L_d \rangle$. By using Einstein's relation ($\sigma \propto d$, where σ is electrical conductivity and d is a diffusion coefficient), Sahimi (1994) and Wong (1999) obtained the same result. Others, e.g., Fatt (1959), Klinkenberg (1951) and Garrouch et al. (2001), found experimentally that τ_e and τ_d were almost the same. Nonetheless, sometimes measurements of electrical conductivity do not correctly predict diffusion (see, e.g., Bezzar and Ghomari, 2013). In such cases, the disparity must stem from secondary phenomena that violate the analogy between diffusion and electrical conductivity, for example, (i) solid-phase conduction, (ii) reactions between the diffusing solute and the solid phase, (iii) a diffuse double layer in which diffusion is slowed by electrokinetic drag while electrical conductivity is enhanced by the higher charge density, or (iv) anion exclusion, which reduces diffusion while increasing electrical conductivity.

UNIFYING CONCEPTS AND CONTRASTS

To put the discussion on a firmer foundation, let us use an abstract, but nevertheless reasonable, model of a pore space, namely a network of interconnected bonds in which each bond represents a pore of a porous medium. Thus, imagine that we have a square network of cylindrical pores, with pore radii assigned at random from a given statistical distribution. Consider various types of transport, e.g., hydraulic or electrical, through the pores. Let the conductance g of an individual pore be proportional to its radius to the n th power ($g \propto r^n$, where r is the pore radius). The geometric tortuosity would force a choice of $n = 0$, while electrical and diffusive tortuosities require $n = 2$ and hydraulic $n = 4$ (Ewing and Hunt, 2006; Hunt and Ewing, 2009). As the exponent increases, flow is increasingly concentrated in fewer pathways, which become increasingly tortuous. Therefore, theoretically, $\tau_g < \tau_e < \tau_h$. We use a Wheatstone bridge configuration, a simple model shown in Fig. 2, to investigate this proposition. The connection across the

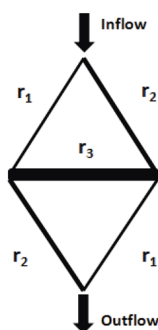


Fig. 2. The Wheatstone bridge configuration in which r_1 , r_2 , and r_3 are pore radii. Note that r_3 is infinitely large. This simple model demonstrates that geometric tortuosity $\tau_g < \text{electrical tortuosity } \tau_e$ (or diffusive tortuosity τ_d) $< \text{hydraulic tortuosity } \tau_h$. More details are given in Table 1.

middle in a Wheatstone bridge is essentially a short circuit, corresponding in fluid flow to a pore of infinite radius. We define the tortuosity relative to the shortest possible path across the system, which is twice the length of an individual element. If one wanted to define the tortuosity relative to a straight-line length through the middle of the bridge, one would simply divide each of our values by the ratio $\sqrt{3}/2 \approx 0.866$; such a constant factor would have no influence on our conclusions.

Our findings are shown in Table 1. Note that for the case of no pore size variability, the geometric tortuosity is identical to both flux-based calculations. The hydraulic tortuosity increases more rapidly with increasing pore-size variability than does the electrical tortuosity, but in the limit of infinite pore-size variability, both reach the same maximum of 1.5. This is analogous to previous findings (Ewing and Hunt, 2006) for the saturation dependence of hydraulic conductivity $K(\theta)$ and electrical conductivity $\sigma(\theta)$, both of which follow universal results from percolation theory for homogeneous media (analogous to Kozeny–Carman and Archie’s law); with increasing heterogeneity, however, the hydraulic conductivity develops a nonuniversal behavior earlier. Finally, in the limit of high heterogeneity, both conductivities exhibited nonuniversal behaviors predicted by relationship $K(\theta) = r_c^2 \sigma(\theta)$ (where r_c is the critical pore radius for percolation, the smallest pore in the connected path on the sample-spanning [percolating] cluster that has the largest possible value of the smallest pore and controls the flow as a bottleneck) (Friedman and Seaton, 1998), implying that both forms of transport occurred along precisely the same pathways.

At this point, we require some perspective on the expected and possible values of tortuosity. All conceptual formulations of tortuosity should be consistent with a value of 1 in the limit that porosity is 1. In contrast, at zero porosity, the system cannot percolate so defining a tortuosity would be meaningless; there is no path through the pore space across the system. In most cases, one expects the percolation transition (Hunt and Ewing, 2009) at a small but finite porosity value called the *percolation threshold*, ϕ_c ; tortuosity is a meaningful quantity only for $\phi > \phi_c$.

As a further consideration, the mathematics of fractals allows path lengths and their associated tortuosities to diverge (meaning path lengths get infinitely large), at least in principle. Consider the Wheatcraft and Tyler (1988) fractal curve length definition $L(\epsilon) = L_s^{D_T} \epsilon^{1-D_T}$, where D_T is the tortuosity fractal dimension, varying in the range 1 to 3 in a three-dimensional medium, ϵ is the scale of measurement, L_s is the straight-line length, and $L(\epsilon)$ is the total length of a curve or path. The length of the pathway approaches infinity either when the system size is infinitely large or as the scale of measurement $\epsilon \rightarrow 0$. In fact, since the only connected pathways at the percolation threshold are fractal in nature, a calculation such as that presented by Wheatcraft and Tyler (1988) should always apply at ϕ_c . In practice, however, the scale of measurement ϵ does not approach zero, nor does the sample size L_s approach infinity. Thus, while the tortuosity may become large in this limit, it remains bounded.

Table 1. Different types of tortuosity for two thread- and flux-based approaches and different radius ratios for the Wheatstone bridge equivalent pore network model presented in Fig. 2.

Radius ratio (r_2/r_1)	Thread-based	Flux-based	
	Geometrical	Electrical	Hydraulic
1	1.25	1.25	1.25
2	1.25	1.30	1.44
10	1.25	1.49	1.4999
Infinite	1.5	1.5	1.5

SPECIFIC TORTUOSITY MODELS FOR SATURATED POROUS MEDIA

Geometrical Models

Models of geometric tortuosity have been commonly developed based on geometric (e.g., particle size, shape, and arrangement) properties, and also topological (e.g., the dimensionality and connectivity of the network, the coordination number—the number of pores connected to each other at each intersection—etc.) properties of a porous medium. These models, which approximately describe the geometric characteristics of the flow path, are generally applicable only to specific artificial porous media. For example, Yu and Li (2004) proposed a tortuosity model for a porous medium consisting of two-dimensional square solid particles, given by

$$\tau_g|_{\text{sat}} = \frac{1}{2} \left[1 + \frac{1}{2} \sqrt{1-\phi} + \frac{\sqrt{(1-\sqrt{1-\phi})^2 + (1-\phi)/4}}{1-\sqrt{1-\phi}} \right] \quad [7]$$

This model allows flow only between square particles in an equilateral-triangle arrangement with either (i) the flow path changing with a specific angle that depends on the particle size and pore space, or (ii) the streamline direction changes are limited to multiples of 90° . However ingenious its derivation, a porous medium with such underlying assumptions is far from any natural porous material.

Yun et al. (2006) developed geometric tortuosity models in three-dimensional porous media consisting of spherical, cubic, or rectangular solid particles. The particles were positioned in the form of equilateral triangle and square unit cell arrangements; however, Yun et al. (2006) did not state explicitly how the unit cells were arranged in a network, i.e., whether the unit cells formed parallel planes or more complicated structures. Because the flow vector was implied to lie within the plane formed by the triangles (or squares), our interpretation is that the unit cells were arranged in a single sheet, and introducing additional sheets would not affect the final results. Note that in their model using spherical particles for tortuosity, any $\phi < 0.476$ (calculated from their Eq. [8]) and 0.395 (calculated from their Eq. [5]) leads to a case of overlapping spheres in the square and triangular unit cells, respectively. But natural granular porous media are rarely monosized, and may have a porosity as small as 0.3 to 0.4 (gravel), 0.3 to 0.35 (gravel and sand), 0.1 to 0.2 (sandstone), or 0.01 to 0.1 (shale or limestone), as given by Bear (1972, p. 46). A poorly

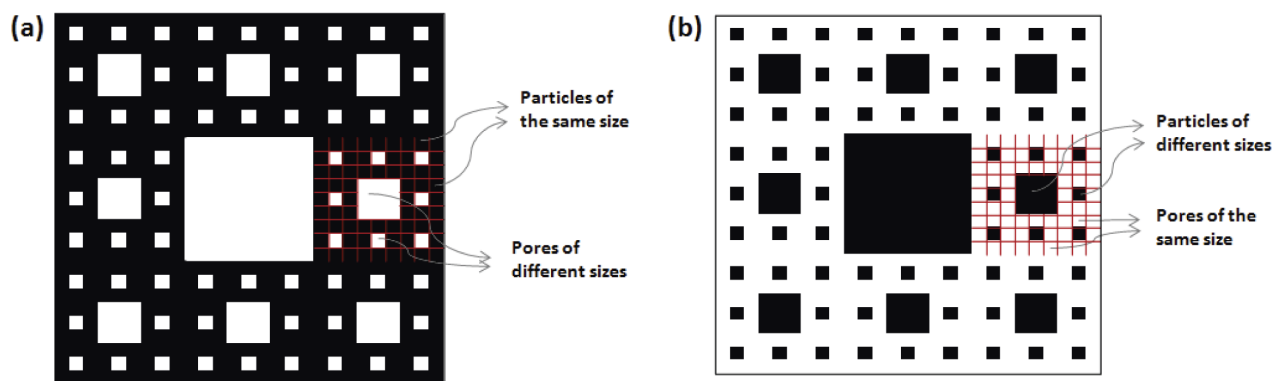


Fig. 3. (a) A solid fractal model constructed of particles of the same size but pores of different sizes, and (b) a pore fractal model built up of particles of different sizes but pores of the same size. Note that the solid matrix is shown in black, while pores are white.

sorted material and a cubic arrangement of spherical grains of two sizes may have porosity values of 0.17 and 0.125, respectively (Bear, 1972, p. 47). Clearly, models based on monosize particles cannot represent natural sedimentary materials. Nonetheless, the results of Yun et al. (2006) broadly agree with the results of Wyllie and Gregory (1955) and Comiti and Renaud (1989), described below (Eq. [11]) with a logarithmic function of porosity.

Feng and Yu (2007) defined tortuosity in porous media by means of fractal geometry. They used the Wheatcraft and Tyler (1988) fractal curve length definition and derived a fractal tortuosity model assuming a hierarchical structure in a saturated porous medium. Their model takes the form:

$$\tau_g \Big|_{\text{sat}} = \int_{r_{\min}}^{r_{\max}} \tau(r) f(r) dr \quad [8]$$

$$\approx \frac{D}{D + D_T - 1} \left(\frac{L_s}{r_{\min}} \right)^{D_T - 1}$$

where $\tau(r) = (L_s/r)^{D_T - 1}$ is the tortuosity for a pore pathway with radius r , with L_s being the straight-line length, $f(r) = Dr_{\min}^D r^{-1-D}$ is the pore size probability density function, D is the fractal dimension of the pore space, $r_{\min}$ and $r_{\max}$ are the smallest and largest pore radii, respectively; and D_T is the tortuosity fractal dimension, restricted to the interval $1 < D_T < E$ (where E is the Euclidean dimension). The Feng and Yu (2007) approach (Eq. [8]) is restricted to saturated media because the full range of the pore sizes is used. Because Eq. [8] is based on concepts of geometric fractals applied to the structure of the medium itself, we argue that applying Eq. [8] to unsaturated conditions has no correspondence to the real flow path in porous media: it gives a cumulative pore length rather than the distance actually travelled by the fluid. Additionally, in Eq. [8] Feng and Yu (2007) assumed that pore length is related to pore size by the Wheatcraft and Tyler (1988) fractal curve length given above, but this means that a pore's length $L(r)$ is inversely proportional to its radius r . Although Fatt (1956) considered this to be the case, in a self-similar fractal medium $L(r)$ should be proportional to r (Ghanbarian-Alavijeh and Hunt, 2012b). This direct proportionality is in agreement with the Mualem

(1976) approach, as well as with experiments on fracture networks (Bonnet et al., 2001).

Lanfrey et al. (2010) assumed that each tortuous path is represented by a sinuous tube with constant cross-sectional area and perimeter, and developed a theoretical model for the tortuosity of a fixed bed of randomly packed identical particles:

$$\tau_g \Big|_{\text{sat}} = 1.23 \frac{(1-\phi)^{4/3}}{\xi^2 \phi} \quad [9]$$

where ξ is the sphericity (or roundness) factor, equal to 1 for spheres and < 1 for nonspherical particles (Lanfrey et al., 2010). A shortcoming of this model is that as $\phi \rightarrow 1$, $\tau_g \rightarrow 0$; this is physically unrealistic because in this limit the tortuosity must be 1, and by definition the tortuosity cannot be < 1 .

Recently, Li and Yu (2011) developed a tortuosity model based on the hierarchical structure of the deterministic Sierpinski carpet; however, heterogeneous soils and fractured networks are better analyzed as statistical (random) fractals rather than deterministic Sierpinski (or Menger sponge) models. The Sierpinski carpet used by Li and Yu (2011) is a pore fractal model that includes particles of different sizes and pores all of the same size at each iteration (Rieu and Perrier, 1997) (Fig. 3b). Although in the pore fractal model (Fig. 3b) only the pore phase is fractal (the solid phase is not *geometrically* fractal), the solid phase is scaled with a power-law function and the same exponent and fractal dimension scaling the fractal pore phase (Rieu and Perrier, 1997). In contrast to the pore fractal model, a solid fractal model has particles of the same size and pores of a range of discrete sizes at each iteration (Fig. 3a). The Li and Yu (2011) model yields

$$\tau_g \Big|_{\text{sat}} = \left(\frac{19}{18} \right)^{\ln(\phi)/\ln(8/9)} \quad [10]$$

Characteristics, descriptions, and comments on different geometric models are summarized in Table 2. Comparing the predicted saturated geometric tortuosity $\tau_g \Big|_{\text{sat}}$ (Fig. 4) from the Yu and Li (2004), Lanfrey et al. (2010), and Li and Yu (2011) models indicates that the power-law form (Eq. [9]) diverges more sharply than the Yu and Li (2004) model (Eq. [7]). Equations [7], [9] and [10] all yield $\tau_g \Big|_{\text{sat}} \rightarrow \infty$ in the limit of zero

Table 2. The geometric tortuosity models and their characteristics proposed in the literature. Note that concavity is up for all models. See also the equations mentioned for variable definitions and details.

Eq.	Input quantities†	Threshold	Dimensions	As $\phi \rightarrow 1$, $\tau_g _{\text{sat}}^{\ddagger} \rightarrow$	Divergence for $\phi \rightarrow 0$	Remarks	Description	Reference
[7]	ϕ	0	two	1	power	square particles	analytical	Yu and Li (2004)
[8]	$D, D_T, L_s, r_{\min}$	–	two, three	1	–	fractal media	analytical	Feng and Yu (2007)
[9]	ϕ, ξ	0	three	0	power§	sinuous channel	analytical	Lanfrey et al. (2010)
[10]	ϕ	0	two	1	power¶	Sierpinski carpet	analytical	Li and Yu (2011)
–#	ϕ	–	three	NA††	none	spherical particles	analytical	Yun et al. (2006)
–#	ϕ	0	three	1	power§	cubic particles	analytical	Yun et al. (2006)
–#	ϕ	0	three	1	power¶	plate-like particles	analytical	Yun et al. (2006)

† ϕ , porosity; D , pore space fractal dimension; D_T , tortuosity fractal dimension; L_s , straight-line length; ξ , sphericity factor.

‡ Saturated geometric tortuosity.

§ Power equal to -1 .

¶ Nonintegral power.

Corresponding equation is not presented in the text.

†† NA, not available.

porosity. Figure 4 also shows that the Yu and Li (2004) and Li and Yu (2011) models predict the tortuosity similarly for porosities >0.60 . Equation [8] is not written explicitly in terms of the porosity, so we have not graphically compared it with the other models.

Hydraulic Conductivity Models

Perhaps the first rough estimate of hydraulic tortuosity was proposed by Carman (1937) as $\langle L_h \rangle / L_s = \sec(\alpha)$, where α is the average angle between the flow pathway and the apparent direction of flow. For cubic packing of equal-sized spheres, $\alpha \approx 45^\circ$; thus $\langle L_h \rangle / L_s \approx \sqrt{2}$ and $\tau_h \approx 2$ (Carman, 1937). A value of $\tau_h = 3$ was then proposed by Dullien (1975) for a cubic network built up from the parallel capillary model (more references given in Dullien, 1979, p. 225).

One of the most invoked models of tortuosity in the literature is a logarithmic function of porosity, given by

$$\tau_h|_{\text{sat}} = 1 - P \ln(\phi) \quad [11]$$

where P is a constant found experimentally to be 1.6 (Pech, 1984) for wood chips (cited in Comiti and Renaud, 1989). Comiti and Renaud (1989) found a range of 0.86 to 3.2 for plates with different height/side ratios. Using the results of Molerus (1980), Mauret and Renaud (1997) reported that a value of $P = 0.49$ is appropriate for a capillary model of high-porosity beds composed of spheres and fibers.

Mota et al. (2001) proposed an empirical tortuosity–porosity power law for binary mixtures of spherical particles:

$$\tau_h|_{\text{sat}} = \phi^{-\beta} \quad [12]$$

where $\beta = 0.4$ was found by measuring the conductivity of porous media. A problem with empirical models is that by definition the coefficient(s), in this case the exponent, must be determined experimentally or numerically; however, two porous media with the same porosity may have different transport path lengths, as demonstrated by Valdés-Parada et al. (2011), so tortuosity can-

not be a function of porosity only. Consequently, the exponent β in Eq. [12] cannot be universal, at least not without replacement of the porosity by some more general function, i.e., inclusion of a threshold. The variability of β is probably due to variations in pore connectivity, but this has not been examined. A similar power law, but in terms of water content rather than porosity, has been widely used in unsaturated hydraulic conductivity models and is discussed below in that context.

In addition to empirical equations, analytical models have also been developed to describe tortuosity in porous media. Using volume-averaging concepts, Du Plessis and Masliyah (1991) derived an analytical model for isotropic granular media:

$$\tau_h|_{\text{sat}} = \frac{\phi}{1 - (1 - \phi)^{2/3}} \quad [13]$$

Note that the saturated hydraulic tortuosity $\tau_h|_{\text{sat}}$ in Eq. [13] ranges between 1 and 1.5 and does not include a percolation threshold, or the critical porosity for macroscopic connectivity, in the medium.

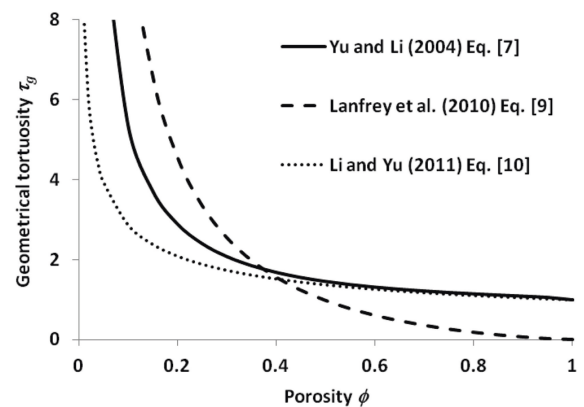


Fig. 4. Comparison of the prediction of geometric tortuosity τ_g using the Yu and Li (2004) model, Eq. [7], the Lanfrey et al. (2010) model, Eq. [9], and the Li and Yu (2011) model, Eq. [10], in a saturated porous medium.

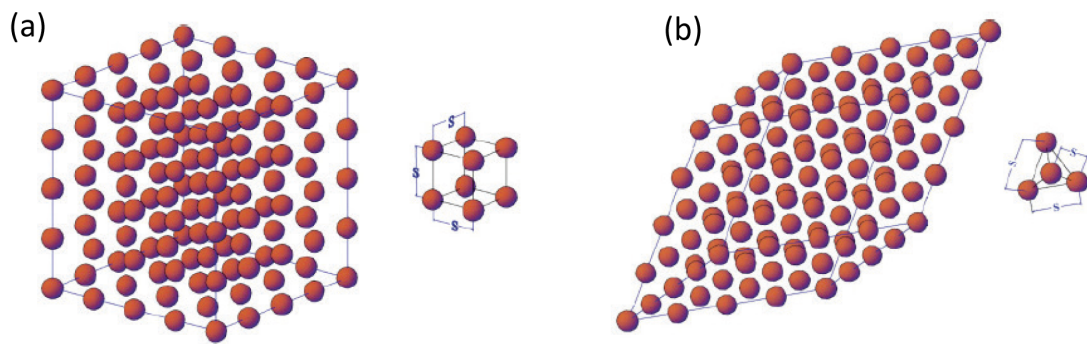


Fig. 5. (a) Cubic and (b) tetrahedral arrays of monosized spheres used in the Ahmadi et al. (2011) hydraulic tortuosity model (after Ahmadi et al., 2011).

Ahmadi et al. (2011) also proposed an analytical derivation of tortuosity for monosized spheres using a volume-averaging approach:

$$\tau_h|_{\text{sat}} = \sqrt{\frac{2\phi}{3[1-B(1-\phi)^{2/3}]} + \frac{1}{3}} \quad [14]$$

in which B is a constant equal to 1.209 for cubic packings (Fig. 5a, after Ahmadi et al., 2011) and 1.108 for tetrahedral (Fig. 5b, after Ahmadi et al., 2011). Note that Eq. [14] diverges as the $-1/2$ power of $\phi - \phi_t$ at the threshold porosity $\phi_t = 1 - B^{-3/2}$. For cubic and tetrahedral packings, the threshold porosity ϕ_t as calculated from Eq. [14] would therefore be 0.248 and 0.143, respectively, close to 0.2488 and 0.15, the bond percolation threshold values reported by Stauffer and Aharony (1994), Sahimi (1994), and Renault (1991). Note that we have respected the original authors' choice in regard to the significant figures reported.

In the approach of Dullien (1979, p. 225), as followed by Hager et al. (1997), a theoretical model was proposed to calculate $\tau_h|_{\text{sat}}$ from the intrinsic permeability, porosity, and pore size distribution. This model yields

$$\tau_h|_{\text{sat}} = \frac{\phi}{8k} \int_{r_{\min}}^{r_{\max}} r^2 f(r) dr \quad [15]$$

where k is the intrinsic permeability, $f(r)$ is the pore size distribution function, and $r_{\min}$ and $r_{\max}$ are the smallest and largest pore radii of the medium, respectively. Because k is typically assumed to be given by an arithmetic average of the individual pore conductivities (i.e., the integral in Eq. [15]), this definition of tortuosity amounts to saying that tortuosity defines the discrepancy between the calculated value of k and experiment; however, the assumption regarding the arithmetic average is erroneous (Kirkpatrick, 1973; Sahimi, 1994; Stauffer and Aharony, 1994; Milton, 2002; Hunt and Ewing, 2009; Sahimi, 2011), and one could less charitably refer to tortuosity so defined as a highly refined fudge factor.

Numerical methods have also been applied to model the hydraulic tortuosity in saturated porous media using lattice-gas cellular automata (Zhang and Knackstedt, 1995; Koponen et al., 1996, 1997) and lattice-Boltzmann models (Matyka et al., 2008;

Duda et al., 2011); however, hydraulic tortuosity modeling using numerical simulations includes many hidden problems that may lead to incorrect conclusions. For example, splitting and merging of a fluid stream as it hits a solid particle causes discontinuities in streamlines. Because the location of each streamline is a priori unknown, the discontinuity in a streamline may lead to ill-conditioned results (Matyka et al., 2008).

Using a lattice-gas cellular automaton method, Koponen et al. (1996) investigated flow in a two-dimensional porous medium, constructed by randomly distributed solid rectangles of equal size and with unrestricted overlap. They found that the hydraulic tortuosity was linearly related to the porosity:

$$\tau_h|_{\text{sat}} = 1 + 0.8(1 - \phi) \quad [16]$$

Equation [16] indicates that when $\phi \rightarrow 1$, the hydraulic tortuosity approaches 1 ($\tau_h \rightarrow 1$). As $\phi \rightarrow 0$, however, $\tau_h \rightarrow 1.8$.

Koponen et al. (1997) used lattice-gas cellular automata and numerically modeled the hydraulic tortuosity for flow of a Newtonian incompressible fluid in a two-dimensional porous medium with a threshold porosity of 0.33 ($\phi_t = 0.33$). In contrast to Koponen et al. (1996), Koponen et al. (1997) used periodic boundary conditions in the lattice to decrease the finite-size and boundary effects. In addition, to include the effect of nonconducting pores, they considered the effective porosity, the ratio of the volume of the conducting pores to the total volume (ϕ_{eff}), rather than the total porosity (ϕ) of the medium. Koponen et al. (1997) proposed the porosity-dependent tortuosity function

$$\tau_h|_{\text{sat}} = 1 + a \frac{1 - \phi}{(\phi - \phi_t)^\gamma} \quad [17]$$

where $a = 0.65$ and $\gamma = 0.19$ are constant parameters, ϕ_t is the threshold porosity, and ϕ is the total porosity. Note that as $\phi \rightarrow 1$, the saturated tortuosity approaches 1. Note also that a power law with a small exponent is often difficult to distinguish from a logarithmic law.

Matyka et al. (2008) used a lattice-Boltzmann model to simulate the porosity dependence of tortuosity for laminar fluid flow in a two-dimensional porous medium composed of freely overlapping solid squares that had a percolation threshold of 0.367. In contrast to Koponen et al. (1996), who weighted the

tortuosities of each streamline by local fluid velocities, Matyka et al. (2008) weighted with local fluxes and found a value of 0.77 for the constant P in Eq. [11].

Recently, Duda et al. (2011) presented a brief overview of the aforementioned three numerical approaches, namely, those of Zhang and Knackstedt (1995), Koponen et al. (1996), and Matyka et al. (2008), as well as a new method that enables calculation of the hydraulic tortuosity without the need to determine streamlines. Determining streamlines is often an ill-conditioned numerical problem, especially if only the approximate fluid velocity field is available. Duda et al. (2011) assumed a percolation threshold of 0.367 and provided an analysis of their numerical results as a function of system size that would support a divergence of the tortuosity at the percolation threshold. Because they were particularly interested in the analytical limit $\phi \rightarrow 1$, they concentrated their analysis of the porosity dependence in the region $\phi > 0.8$, where they found that the empirical function

$$\tau_h|_{\text{sat}} = 1 + \chi \sqrt{1 - \phi} \quad [18]$$

best described the results of their simulations. Note that the value of χ was not reported by Duda et al. (2011).

The saturated hydraulic tortuosity models (Table 3) make some similar predictions (Fig. 6). For example, Eq. [11] with $P = 0.49$, Eq. [12] with $\beta = 0.4$, and Eq. [14] with $B = 1.209$ all predict similar $\tau_h|_{\text{sat}}$ for $\phi > 0.4$. Differences are greater for $\phi < 0.4$. In particular, $\tau_h|_{\text{sat}}$ predicted by Eq. [14] with $B = 1.209$ increases rapidly as ϕ decreases, and diverges at $\phi = 0.248$. Figure 6 also indicates that the shape of the saturated hydraulic tortuosity predicted by the Du Plessis and Masliyah (1991) model (Eq. [13]) is convex, whereas all other models predicted a concave shape for $\tau_h|_{\text{sat}}$. In fact, given the suggested bounds of $\tau \rightarrow 1$ for $\phi \rightarrow 1$, and $\tau \rightarrow \infty$ for $\phi \rightarrow 0$, the concave shape is required. Experimental (Delgado, 2006; Barrande et al., 2007) and numerical (Koponen et al., 1997; Matyka et al., 2008; Duda et al., 2011) results also support a concave curve for this relationship.

Electrical Conductivity Models

Based on the solution of Laplace's equation for steady-state conduction, Maxwell (1873) proposed a formula for the electrical conductivity of a conducting medium having a dilute suspension of nonconducting spheres, indicating that the electrical tortuosity, τ_e , satisfies

$$\tau_e|_{\text{sat}} = 1 + \frac{1}{2}(1 - \phi) \quad [19]$$

Equation [19] implies that as $\phi \rightarrow 0$, $\tau_e|_{\text{sat}} \rightarrow 1.5$. Note that Maxwell's derivation was for the dilute limit (i.e., when the volume fraction of the nonconducting particles is small) and was never intended to be extrapolated to zero porosity.

If the empirical Archie's law ($F = \phi^{-m}$) (Archie, 1942) is combined with Eq. [3], the resulting equation would be

$$\tau_e|_{\text{sat}} = \phi^{1-m}, \quad m \geq 1 \quad [20]$$

Table 3. The hydraulic tortuosity models and their characteristics proposed in the literature. See also the equations mentioned for variable definitions and details.

Eq.	Input quantities†	Adjustable parameters	Threshold	Dimensions	Concavity	As $\phi \rightarrow 1$, $\tau_h _{\text{sat}} \rightarrow$	Divergence for $\phi \rightarrow 0$ or ϕ_t	Remarks	Description	Reference
[11]	ϕ	$P = 1.6$	0	three	up	1	logarithmic	wood chips	empirical	Pech (1984)
[11]	ϕ	$P = 0.86-3.2$	0	three	up	1	logarithmic	platy particles	empirical	Comiti and Renaud (1989)
[11]	ϕ	$P = 0.49$	0	three	up	1	logarithmic	high-porosity beds	empirical	Maurel and Renaud (1997)
[12]	ϕ	$\beta = 0.40$	0	three	up	1	power	binary mixtures of spherical particles	empirical	Mota et al. (2001)
[13]	ϕ	—	—	three	down	1	none	isotropic granular media	analytical	Du Plessis and Masliyah (1991)
[14]	$\phi, B = 1.209$	—	0.248	three	up	1	power§	cubic packing	analytical	Ahmadi et al. (2011)
[14]	$\phi, B = 1.108$	—	0.143	three	up	1	power§	tetrahedral packing	analytical	Ahmadi et al. (2011)
[15]	$\phi, k, f(r)$	—	—	three	up	—	none	—	analytical	Dullien (1979)
[16]	ϕ	—	—	two	none	1	none	randomly distributed squares with unrestricted overlap	numerical	Koponen et al. (1996)
[17]	ϕ, ϕ_t	$a = 0.65, \gamma = 0.19$	0.33	two	up	1	power¶	randomly distributed squares with unrestricted overlap	numerical	Koponen et al. (1997)
[11]	ϕ	$P = 0.77$	0.367	two	up	1	logarithmic	freely overlapping squares	numerical	Matyka et al. (2008)
[18]	ϕ	χ	0.367	two	up	1	power§	porosity value > 0.8	numerical	Duda et al. (2011)

† ϕ , porosity; B , constant coefficient; k , intrinsic permeability; $f(r)$, pore size distribution function; ϕ_t , threshold porosity.

‡ Saturated hydraulic tortuosity.

§ Nonzero threshold.

¶ Nonintegral power.

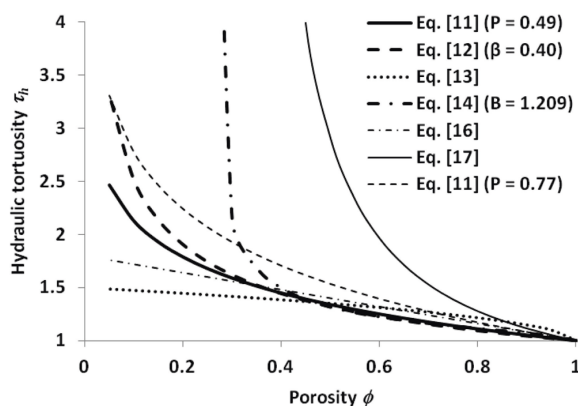


Fig. 6. Comparison of the prediction of hydraulic tortuosity τ_h using the Mauret and Renaud (1997) model, Eq. [11], with constant $P = 0.49$; the Mota et al. (2001) model, Eq. [12], with exponent $\beta = 0.4$; the Du Plessis and Masliyah (1991) model, Eq. [13]; the Ahmadi et al. (2011) model, Eq. [14]; the Koponen et al. (1996) model, Eq. [16]; the Koponen et al. (1997) model, Eq. [17], with percolation threshold $\phi_t = 0.33$; and the Matyka et al. (2008) model, Eq. [11], with $P = 0.77$ in a saturated porous medium.

which is similar to Eq. [12] but with an exponent that generally lies between -3.4 and -0.2 (Friedman, 2005). The larger the m value, the more tortuous the path (Wong, 1999). The exponent m may depend on a variety of factors, as discussed in detail by Sahimi (2003), including the shape and polarization of the particles.

Using Einstein's equation (Einstein, 1905) that expresses the relationship between electrical conductivity and the diffusion coefficient, Coleman and Vassilicos (2008) applied an approach similar to that of Roy and Tarafdar (1997), defined electrical tortuosity as Eq. [3], and consequently found a power-law divergence of τ_e as a function of porosity under saturated conditions:

$$\tau_e|_{\text{sat}} = \phi^{(D_w - 2)/(D_m - E)} \quad [21]$$

where D_w is the random walk fractal dimension, i.e., 2 divided by the exponent that defines the power-law dependence of the mean-square displacement of a diffusant with time ($D_w > 2$; $D_w = 2$ refers to Fickian diffusion), D_m is the mass fractal dimension of the Sierpinski carpet (or the Menger sponge in three dimensions) ($0 < D_m < E$), and E is the Euclidean dimension. One should substitute θ for ϕ to apply Eq. [21] to unsaturated conditions. Coleman and Vassilicos (2008) reported a range of -0.4 to -2 for the exponent in Eq. [21]. Note that electrical tortuosity in Eq. [21] tends to 1 as porosity approaches 1 and diverges at zero porosity. In addition, in Eq. [21] the percolation threshold is zero: the system remains connected down to zero porosity or saturation.

Comparison of Eq. [20] and [21] with Eq. [12] ($\tau_h|_{\text{sat}} = \phi^{-\beta}$) would imply that $\tau_e > \tau_h$ in a saturated medium; however, Eq. [12] was proposed empirically for binary mixtures of spherical particles. Dullien (1979), David (1993), and Zhang and Knackstedt (1995) found $\tau_h > \tau_e$. As we explained above, the geometric, electrical (or diffusive), and hydraulic tortuosities require the pore conductances to be proportional to their radius

to the zeroth, second, and fourth powers, respectively, meaning that the hydraulic flow is concentrated in fewer (and, accordingly, more tortuous) pathways than the electrical (or diffusive) and geometric flows. As a result, one can theoretically assume that $\tau_g < \tau_e (\approx \tau_d) < \tau_h$, as pointed out above.

Using the data presented in Wyllie and Gregory (1955), and calculating electrical tortuosity from Eq. [3], Comiti and Renaud (1989) found $P = 0.41$ for monosized and polydisperse spheres and $P = 0.63$ for cubes in Eq. [11].

Larson and Davis (1982) introduced three distinct definitions of the tortuosity in terms of the electrical conductivity and percolation properties of random conductance networks. These definitions are: (i) the ratio of p , the conducting volume fraction, and the conductivity fraction of the network, the ratio of the conductivity of a lattice with a random fraction p of identical conducting bonds to the conductivity of the same lattice with all bonds conducting, (ii) the ratio of percolation probability—the conducting volume fraction in the sample-spanning cluster—and the conductivity fraction, and (iii) the ratio of the backbone fraction—the conducting volume fraction of in the backbone of the network, i.e., the current-carrying part (some parts are dead-end and carry no current)—and the conductivity fraction of the network. All three definitions yield a tortuosity that diverges at the percolation threshold (critical porosity) and approaches 1 in the limit of unit porosity.

Barrande et al. (2007) measured the porosities and formation factors of porous media composed mostly of spherical particles and calculated the tortuosity using Eq. [3]. They found that Eq. [11] with $P = 0.49$ reported by Mauret and Renaud (1997) was in agreement with their calculated electrical tortuosity in a monosized glass bead medium.

The various electrical tortuosity models (Table 4) give similar predictions at high porosity values (Fig. 7). Specifically, the predicted saturated electrical tortuosity $\tau_e|_{\text{sat}}$ using the Maxwell (1873) model and the Barrande et al. (2007) model (with $P = 0.49$) are quite close for porosity > 0.6 . This result was mathematically anticipated by Weissberg (1963), as we explain below.

Liquid-Phase Diffusion-Based Models

Solute species spreading due to the thermal energy of molecules is called *diffusion*. Net transport of solutes by diffusion occurs in response to spatial differences in concentration. Diffusion may happen in the solid, liquid, and gas phases. It has been asserted that diffusion characteristics in each phase should be unique because the pore geometry and connectivity are different (Moldrup et al., 2001; Flury and Gimmi, 2002). This statement, while apparently obvious, tends to obscure certain unifying characteristics in, e.g., the saturation dependence of the tortuosity in the different phases, which can be predicted using percolation theory (Sahimi, 1993; Hunt and Ewing, 2003). Insofar as descriptions of the air permeability (Ghanbarian-Alavijeh and Hunt, 2012a) and electrical conductivity (Ewing and Hunt, 2006) follow universal scaling from percolation theory, the associated tortuosities should also.

Table 4. The electrical tortuosity models and their characteristics proposed in the literature. For all models, as porosity $\phi \rightarrow 1$, the saturated electrical tortuosity $\tau_e|_{\text{sat}} \rightarrow 1$. See also the equations mentioned for variable definitions and details.

Eq.	Input quantities†	Adjustable parameters	Threshold	Dimensions	Concavity	Divergence for $\phi \rightarrow 0$	Remarks	Description	Reference
[19]	ϕ	–	–	three	none	none	dilute suspension of nonconducting spheres	analytical	Maxwell (1873)
[20]	ϕ	m	0	three	up	power‡	petroleum industry	empirical	Archie (1942)
[21]	ϕ, D_w, D_m, E	–	0	two, three	up	power‡	fractal media	analytical	Coleman and Vassilicos (2008)
[11]	ϕ	$P = 0.41$	0	three	up	logarithmic	spherical particles	empirical	Comiti and Renaud (1989)
[11]	ϕ	$P = 0.63$	0	three	up	logarithmic	cubic particles	empirical	Comiti and Renaud (1989)
[11]	ϕ	$P = 0.49$	0	three	up	logarithmic	monosized glass beads	empirical	Barrande et al. (2007)

† ϕ , porosity; D_w , random walk fractal dimension; D_m , Sierpinski carpet mass fractal dimension; E , Euclidean dimension.

‡ Nonintegral power.

Penman (1940) was a pioneer of the study of gas diffusion in soils. He experimentally found a constant tortuosity coefficient of 0.66; however, it was later acknowledged (see, e.g., Currie, 1960, 1961; Currie and Rose, 1985; Moldrup et al., 2000a, 2000b) that it was unlikely for the diffusive tortuosity coefficient to take on the same value for all types of soils and under all conditions because the tortuous path length should increase as the air-filled pore volume decreases (Hillel, 2004). In this section, we focus on liquid-phase diffusive tortuosity models.

Weissberg (1963) analytically derived a result similar to Eq. [11] with $P = 0.5$ for diffusive tortuosity in porous media, applicable to randomly overlapping spheres of either uniform or nonuniform sizes. Weissberg (1963) also demonstrated that for dilute media where $\phi \rightarrow 1$, Eq. [11] with $P = 0.5$ would be identical to the Maxwell (1873) model (Eq. [19]) up to the first term $(1 - \phi)$.

Equation [11] with $P = 1$ or $\frac{2}{3}$ has been also reported for diffusive tortuosity and various arrangements of cylinders (Tsai and Strieder, 1986). Beeckman (1990) developed an analytical approach to model the nature of highly interconnected pores in a heterogeneous catalyst in two- and three-dimensional systems. His theory is based on a travel process through the catalyst and enables the properties of the pore space to be predicted. Beeckman (1990) found that

$$\tau_d|_{\text{sat}} = \frac{\phi}{1 - (1 - \phi)^{1/3}} \quad [22]$$

As $\phi \rightarrow 0$, $\tau_d|_{\text{sat}} \rightarrow 3$, meaning that $\tau_d|_{\text{sat}}$ varies in the range 1 to 3. Although for networks with the usual range of coordination numbers of four and more, $\tau_h|_{\text{sat}} \approx 3$ is the upper limit for hydraulic tortuosity (Dullien, 1979, p. 225), the range 1 to 3 is unlikely to suffice for describing tortuosity in natural porous media such as soils and rocks. In such materials, hydraulic and electrical conductivities vary across many orders of magnitude; hence the tortuosity will probably also vary widely.

Iversen and Jorgensen (1993) measured diffusion coefficients of SO_4 and CH_4 in seawater and sediments. They determined the diffusive tortuosity τ_d using Eq. [5], finding a linear relationship between τ_d and ϕ similar to Maxwell's formula:

$$\tau_d|_{\text{sat}} = 1 + q(1 - \phi) \quad [23]$$

where $q = 2$ for sandy sediments and 3 for clay and silt sediments.

Delgado (2006) determined the diffusive tortuosity using Eq. [5] after measuring diffusion in free fluid, d_p and in packed beds of silica sand with a narrow range of particle sizes, d_p . The diffusive tortuosity in packed beds agreed well with the empirical model of Mota et al. (2001) (Eq. [12]), with exponent $\beta = 0.4$, and with Eq. [11] with $P = 0.5$ (Weissberg, 1963); it also agreed with the theoretical model proposed by Yun et al. (2005) for the case of porosities > 0.4 .

The characteristics of different diffusive tortuosity models are summarized in Table 5. Equation [11] predicts divergence of the tortuosity at a critical porosity (in particular $\phi = 0$), but its logarithmic form diverges more slowly than power laws do (Fig. 8). In many of the comparisons with simulations, the tortuosity varies over less than an order of magnitude, even when a critical porosity is approached, probably providing the chief rationale for using Eq. [11]. Figure 8 also indicates that prediction of $\tau_d|_{\text{sat}}$ using Eq. [11] with $P = 0.5$ is consistent with Eq. [12] with $\beta = 0.4$ as porosity decreases from 1 to 0.35.

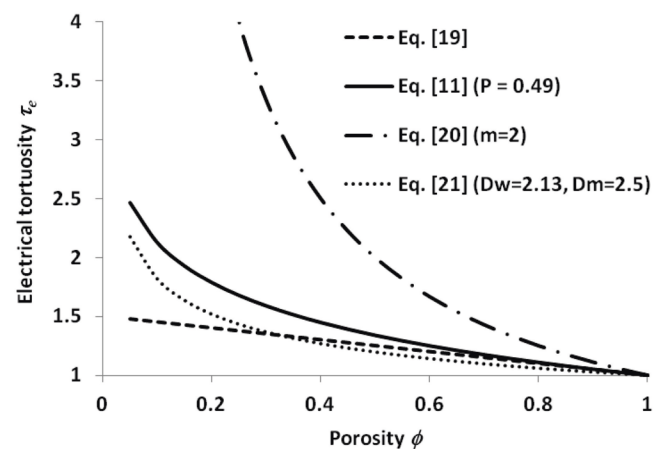


Fig. 7. Comparison of the prediction of electrical tortuosity τ_e using the Maxwell (1873) model, Eq. [19]; Eq. [11] with constant $P = 0.49$; the Archie (1942) model, Eq. [20], with exponent $m = 2$; and the Coleman and Vassilicos (2008) model, Eq. [21], with random walk fractal dimension $D_w = 2.13$ and Sierpinski carpet mass fractal dimension $D_m = 2.50$ in a saturated porous medium.

Table 5. The diffusional tortuosity models and their characteristics proposed in the literature. For all models, as porosity $\phi \rightarrow 1$, the saturated diffusional tortuosity $\tau_e|_{\text{sat}} \rightarrow 1$; all models are in three dimensions and ϕ is the sole input quantity. See also the equations mentioned for variable definitions and details.

Eq.	Adjustable parameters	Threshold	Concavity	Divergence for $\phi \rightarrow 0$	Remarks	Description	Reference
[11]	$P = 0.50$	0	up	logarithmic	randomly overlapping spheres	analytical	Weissberg (1963)
[11]	$P = 1$ or $\frac{2}{3}$	0	up	logarithmic	cylinders	empirical	Tsai and Strieder (1986)
[22]	—	—	down	none	catalyst	analytical	Beeckman (1990)
[23]	$q = 2$ (3)	—	none	none	sandy (clay–silt) sediments	empirical	Iversen and Jorgensen (1993)
[11]	$P = 0.50$	0	up	logarithmic	packed beds of silica sand	empirical	Delgado (2006)
[12]	$\beta = 0.40$	0	up	power†	packed beds of silica sand	empirical	Delgado (2006)

† Nonintegral power.

EXISTING TORTUOSITY MODELS FOR UNSATURATED POROUS MEDIA

To be truly useful, a tortuosity relationship should work across the complete range of saturation, but developing such a model is a formidable task. Many expressions for the saturation-dependent tortuosity are given without specific reference to the property addressed. Consequently, we discuss all saturation-dependent expressions together: geometric, hydraulic, electrical, and diffusive. Where the researchers have made specific mention of the form of the tortuosity intended, our notation will indicate their choice.

Fatt and Dykstra (1951) proposed an empirical microscopic representation for the hydraulic tortuosity τ_h as a function of pore size

$$\tau_{hr} = \frac{\tau_h(r)}{\tau_h|_{\text{sat}}} = \left(\frac{r_{\text{max}}}{r} \right)^j \quad [24]$$

where τ_{hr} is the relative hydraulic tortuosity (the subscript r denotes relative), $\tau_h|_{\text{sat}}$ is the hydraulic tortuosity of the saturated medium, $\tau_h(r)$ is the hydraulic tortuosity of the unsaturated medium, r is the radius of the largest water-filled pore in the unsaturated medium, and r_{max} is the medium's largest pore radius. The empirical exponent j was estimated to be 0.5 by Fatt

and Dykstra (1951), but they found that j values varied in soils. Specifically, they mentioned that j tended to take on values $< \frac{1}{2}$ for Basal Tuscaloosa sand and equal to $\frac{1}{2}$ for Gatchell sand. They also found that for some types of sand j values $> \frac{1}{2}$ led to more accurate hydraulic conductivity predictions but did not provide upper and lower limits. Fatt and Dykstra (1951) may have been the first to use this power-law formulation for tortuosity modeling in porous media.

Burdine (1953) also presented a relative hydraulic tortuosity model as a function of effective saturation (cited in Corey, 1994):

$$\tau_{hr} = \frac{\tau_h(\theta)}{\tau_h|_{\text{sat}}} = S_e^{-\lambda} \quad [25]$$

where effective saturation $S_e = (\theta - \theta_r)/(\phi - \theta_r)$ in which θ is water content, θ_r is residual water content (the volume fraction of water that remains in a pore space after displacement because it is disconnected), ϕ is porosity, and λ is a constant suggested to be equal to 2. Mualem (1976) proposed a similar power function, but in his unsaturated hydraulic conductivity equation, the factor $S_e^{-\lambda}$ (with $\lambda = 0.5$) was intended to correct simultaneously for both pore correlation and flow path tortuosity, with no attempt to distinguish between the two. Equation [25] (and also Eq. [24], in the case that r is given by the Tyler and Wheatcraft [1990] model) is consistent with a power-law divergence of the tortuosity as saturation approaches a critical value (the residual water content). Fitting the Mualem–van Genuchten (Mualem, 1976; van Genuchten, 1980) hydraulic conductivity function to measured unsaturated conductivity values, Schaap and Leij (2000) and Shinomiya et al. (2001) found exponents -1 and -0.77 , respectively. Exponents less than zero, however, result in $\tau_h(\theta)/\tau_h|_{\text{sat}} < 1$, which is physically nonsensical: it implies that removing connections (by draining pores) makes flow less tortuous rather than more.

The widely used Millington and Quirk (1961) saturation-dependent tortuosity model is

$$\tau_h(\theta) = \frac{\phi^2}{\theta^{10/3}} = \left(\frac{\theta}{\phi} \right)^{-2} \theta^{-4/3} \quad [26]$$

Although Millington and Quirk (1961) derived Eq. [26] to predict unsaturated hydraulic conductivity, they also proposed

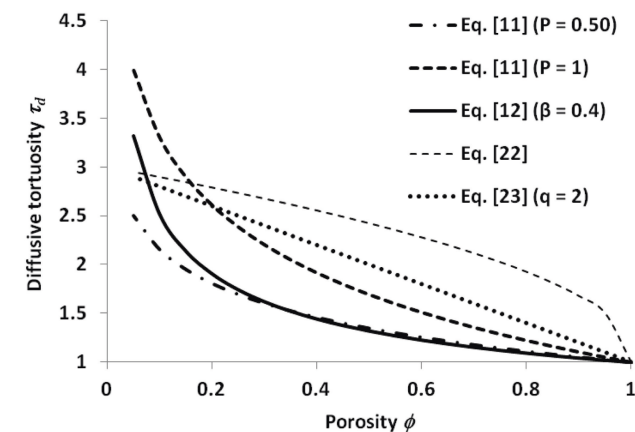


Fig. 8. Comparison of the prediction of diffusive tortuosity τ_d using the Weissberg (1963) model, Eq. [11], with constant $P = 0.50$; the Tsai and Strieder (1986) model, Eq. [11], with $P = 1$; the Delgado (2006) model, Eq. [12], with exponent $\beta = 0.40$; the Beeckman (1990) model, Eq. [22]; and the Iversen and Jorgensen (1993) model, Eq. [23], with soil texture related constant $q = 2$ in a saturated porous medium.

Table 6. Tortuosity models and their characteristics proposed for unsaturated media in the literature. For all models, concavity is up and as the volumetric water content $\theta \rightarrow$ porosity ϕ , relative tortuosity $\tau_r \rightarrow 1$. See also the equations mentioned for variable definitions and details.

Eq.	Input variablest	Adjustable parameters	Threshold	Dimensions	Divergence for $\phi \rightarrow 0$ or θ_r	Remarks	Description	Reference
[24]	$r, r_{\max}$	$j = 0.5$	–	three	power	hydraulic tortuosity	empirical	Fatt and Dykstra (1951)
[25]	θ, θ_r, ϕ	$\lambda = 2$	θ_r	three	power	hydraulic tortuosity	empirical	Burdine (1953)
[25]	θ, θ_r, ϕ	$\lambda = 0.5$	θ_r	three	power	hydraulic tortuosity	empirical	Mualem (1976)
[25]	θ, θ_r, ϕ	$\lambda = -1$	θ_r	three	power	hydraulic tortuosity	empirical	Schaap and Leij (2000)
[25]	θ, θ_r, ϕ	$\lambda = -0.77$	θ_r	three	power	hydraulic tortuosity	empirical	Shinomiya et al. (2001)
[27]	θ, ϕ	–	0	three	power	hydraulic tortuosity	theoretical	Millington and Quirk (1961)
[29]	$\theta, \phi, D_w, D_m, E$	–	0	two, three	power	electrical tortuosity	analytical	Coleman and Vassilicos (2008)

† r , radius of the largest water-filled pore in the unsaturated medium; $r_{\max}$, medium's largest pore radius; θ , volumetric water content; θ_r , volumetric residual water content; ϕ , porosity; D_w , random walk fractal dimension; D_m , Sierpinski carpet mass fractal dimension; E , Euclidean dimension.

its use to estimate diffusion in unconsolidated media, implying that hydraulic and diffusive tortuosities are equivalent. Equation [26], with tortuosity diverging as the water content approaches zero, has been widely used to estimate the ratio d_p/d_f in either gas (replacing θ by ϵ) or liquid phases (see, e.g., Moldrup et al., 2000a, 2003; Hamamoto et al., 2012).

If the medium is saturated, Eq. [26] reduces to $\tau_{h|sat} = \phi^{-4/3}$, which differs markedly from the Mota et al. (2001) and Delgado (2006) results (Eq. [12] with $\beta = 0.4$). Dividing Eq. [26] by the saturated tortuosity $\tau_{h|sat} = \phi^{-4/3}$, we have

$$\tau_{hr} = \frac{\tau_h(\theta)}{\tau_{h|sat}} = \left(\frac{\theta}{\phi}\right)^{-2} \left(\frac{\theta}{\phi}\right)^{-4/3} \quad [27]$$

Note that in the absence of the second factor, $(\theta/\phi)^{-4/3}$, Eq. [27] would reduce to the Burdine model, Eq. [25].

Coleman and Vassilicos (2008) developed a power-law expression for electrical tortuosity as a function of water content in unsaturated porous media:

$$\tau_e(\theta) = \theta^{(D_w - 2)/(D_m - E)} \quad [28]$$

If Eq. [28] is divided by Eq. [21], the relative electrical tortuosity is given by

$$\tau_{er} = \frac{\tau_e(\theta)}{\tau_{e|sat}} = \left(\frac{\theta}{\phi}\right)^{(D_w - 2)/(D_m - E)} \quad [29]$$

Characteristics of tortuosity models proposed for unsaturated media are summarized in Table 6. Comparison (Fig. 9) shows that the Burdine (1953) model (Eq. [25]) with $\lambda = 2$ and the Millington and Quirk (1961) model (Eq. [27]) predict τ_r values that are one to two orders of magnitude greater at small porosity values than values from the Mualem (1976) model (Eq. [25] with $\lambda = 0.5$). The discrepancy between the Burdine (1953) and Millington and Quirk (1961) models is due to the $(\theta/\phi)^{-4/3}$ factor in the Millington and Quirk (1961) model.

Saripalli et al. (2002) assumed that the ratio of the air–water interfacial area in a real porous medium to that in an idealized porous medium is equal to the square of $\langle L_h \rangle / L_s$ and proposed

a new definition of tortuosity for saturated and unsaturated porous media based on the interfacial area ratio. For unsaturated media, their tortuosity is

$$\tau = \frac{a_{aw}}{a_{aw,0}} \quad [30]$$

where a_{aw} is the air–water interfacial area of the unsaturated medium and $a_{aw,0}$ is the same parameter for the corresponding, idealized capillary bundle. This phenomenological relationship does not fit readily into any useful classification scheme and has been used to estimate both unsaturated hydraulic conductivity and unsaturated diffusion in porous media (Saripalli et al., 2002; Khaleel and Saripalli, 2006; Khaleel, 2008).

TORTUOSITY OF POROUS CATALYSTS

Due to the immense industrial significance of porous catalyst pellets, mass transfer phenomena in catalyst pellets have been intensively studied since the early 1960s. The goal was to estimate the tortuosity or (equivalently) the diffusion coefficient of the diffusing species in a given catalyst (Evans et al., 1961; Scott and Dullien, 1962; Wakao and Smith, 1962, 1964; Johnson and Stewart, 1965). Diffusion is the most significant, and often the

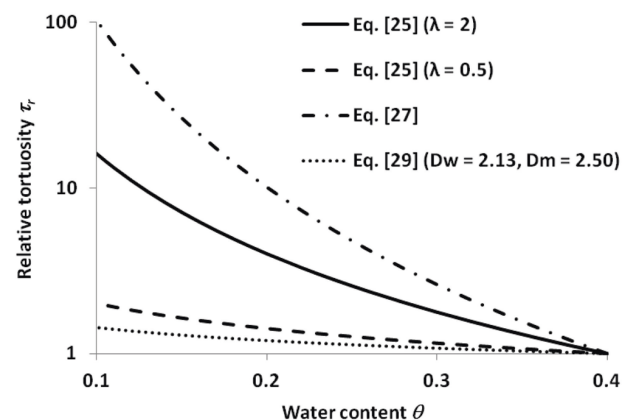


Fig. 9. Comparison of the prediction of relative tortuosity τ_r using the Burdine (1953) model, Eq. [25], with constant $\lambda = 2$; the Mualem (1976) model, Eq. [25], with $\lambda = 0.50$; the Millington and Quirk (1961) model, Eq. [27]; and the Coleman and Vassilicos (2008) model, Eq. [29], with random walk fractal dimension $D_w = 2.13$ and Sierpinski carpet mass fractal dimension $D_m = 2.50$ in an unsaturated porous medium with porosity of 0.4.

only, reactant transport mechanism in porous catalysts, so tortuosity of a catalyst pellet is usually defined via Eq. [6]. The pore structure of most catalyst pellets is as chaotic as that of natural porous material, so advances in modeling pore structures in these two different systems have advanced in parallel.

Similar to early studies of flow through porous media, early studies of the tortuosity of porous catalysts used a one-dimensional model of pore space: a bundle of capillary tubes. Wakao and Smith (1964), for example, proposed a random pore model in a porous catalyst with both micropores and macropores. Diffusion in their model took place through three parallel paths: macropores, micropores, and composite pores consisting of macro- and micropores in series, with diffusion through macropores assumed proportional to ϕ^2 . In a similar spirit, Johnson and Stewart (1965) integrated the diffusion flux over the entire pore size distribution of the pellet, arriving at a tortuosity value of 3. Due to the wide range of pore sizes in many catalyst pellets, many studies considered molecular, Knudsen, and surface diffusion; the derived tortuosity value depended on both molecular and Knudsen diffusion. For example, using a one-dimensional model, Wang and Smith (1983) derived the following equation for d_p and, hence, the tortuosity:

$$d_p = d_f \frac{\phi}{\tau_d} = \frac{1}{\tau_d} \int_{r_{\min}}^{\infty} \frac{f(r)}{(1 - \alpha y_A)/d_A + 1/d_{KA}} dr \quad [31]$$

where $f(r)$ is the pore size distribution of the catalyst, d_A and d_{KA} are the molecular and Knudsen diffusivities of species A , respectively, y_A is the mole fraction of A , α is a constant that depends on the molecular weight of A , and $r_{\min}$ is the lower limit of the pore size (determined by the molecular size). Early work in this field was well summarized by Satterfield (1970), while Carniglia (1986) reviewed the range of tortuosity values found in industrial catalyst pellets. The early pore space models yielded tortuosity values that were roughly consistent with experimental values determined by Eq. [6], but because the models were oversimplified and unrealistic, predictions were generally considered unreliable until more realistic catalyst models were developed in the 1980s.

Sahimi and Tsotsis (1985) were the first to use a pore network model and percolation concepts to model diffusion and reaction in porous catalysts. Pore network models were soon recognized as the most realistic models of catalyst pellets (Sahimi et al., 1990). This recognition enabled researchers to carry out precise and large-scale computations and determine the effective diffusivities and, hence, tortuosity values via Eq. [6]. Advances in instrumentations, such as pulsed-gradient spin echo nuclear magnetic resonance (PGSE-NMR) (Vasenkova and Kortunov, 2005), also enabled precise measurement of the diffusivities. These advances were reviewed by Rigby and Daut (2002).

Typical of catalyst pore network models is that of Hollewand and Gladden (1992), building on the earlier work of Sa-

himi (1988). They varied the connectivity of the pore network, built on a simple cubic lattice and randomized so that the average coordination number could be varied, and showed that the tortuosity was not too sensitive to the connectivity. Most of the tortuosity values that they computed were around 4, remarkably close to what Johnson and Stewart (1965) had computed using a one-dimensional model. But when they used a network with a completely random topology, the resulting tortuosities sometimes differed from those computed on regular networks of the same coordination numbers, although they were still around 4.

Mass transfer in catalyst pellets is always accompanied by chemical reactions, with either linear or nonlinear kinetics. Over the years, a question of great interest has been whether the effective diffusivity of a species in a catalyst is the same with and without reaction. If the two diffusivities are not the same, then the tortuosity value defined by Eq. [6] will be greatly influenced by reaction. The same question is relevant in soil, where transport is typically accompanied by chemical reactions such as sorption and degradation. Experimentally, the subject was controversial for years, with conflicting results abounding in the literature, as summarized by Dadvar and Sahimi (2007). There was thus a real need for precise and large-scale computations to address the issue.

A first step toward addressing this question was taken by Sahimi (1988), who used a variety of network models, an effective-medium approximation, and numerical simulation to study the problem in the presence of a first-order reaction. He reported that effective diffusivities with and without first-order reaction do not differ much unless the pore space is poorly connected (is near its critical porosity or percolation threshold); in that case the two diffusivities (and therefore the tortuosity values) may differ greatly. A more refined simulation study (Hollewand and Gladden, 1992) varied the rate of reaction relative to diffusion and also reported that the tortuosity value is different under reactive and nonreactive conditions if the reaction is first order. Most reactions in industrial applications of porous catalysts—as well as in soil—follow nonlinear kinetics, such as second-order reactions and Michaelis–Menten kinetics. Dadvar and Sahimi (2007) performed extensive numerical simulations using pore network models, as well as analytical analysis, and showed that the effective diffusivities in a pore space in the absence and presence of nonlinear reactions are fundamentally different. The diffusive tortuosity of catalyst pellets and soil therefore depends on reaction kinetics, as well as the other factors we have discussed in this review.

CONCLUSIONS

In the vast literature on porous media, four types of tortuosity—geometric, hydraulic, electrical, and diffusive—have been defined and used. Although the basic concept and definition of tortuosity seem simple, we find that analytical, empirical, and numerical models have been used to estimate different kinds of tortuosity interchangeably, in both saturated and unsaturated media. Because the geometric, electrical (or diffusive), and hydraulic tortuosities require a pore's conductance to be propor-

tional to its radius to the zeroth, second, and fourth power, respectively, we postulated that for higher powers flow is concentrated in fewer pathways; consequently, hydraulic flow becomes more tortuous than electrical (or diffusive) flow, which is more tortuous than geometric “flow”. Using a Wheatstone bridge model, we demonstrated the result that $\tau_g < \tau_e (\approx \tau_d) < \tau_h$. In practice, however, it seems to us that it would be better to restrict definitions of tortuosity to the purely geometric in the sense of path lengths.

Given the remarkable diversity of definitions and concepts of tortuosity in the literature, the subject would benefit from a unifying concept. Experimental inferences regarding tortuosity to date have largely been confounded by inconsistently modeled effects of pore size variability. But some properties, such as the saturation dependence of liquid- and gas-based diffusion, may be related to tortuosity by the universal scaling of percolation theory. Some models concentrate on the geometry of the individual pores for calculating tortuosity, rather than on the geometry of the flow paths. We suggest that approaches that do not recognize this fundamental distinction should be avoided, which means also that experimentally determined values of the tortuosity should not be expected to yield information regarding the structure of the medium.

Many proposed tortuosity functions diverge at a porosity of (or near) zero, while others either do not diverge or diverge only logarithmically. Power-law divergences can probably be related to the relevance of percolation theory. When applicable, scaling relationships from percolation theory, expressed in terms of power laws, can yield useful expressions both for fluid flow properties, such as the air permeability, and for diffusion. In these cases, the property vanishes at the percolation threshold and the power is positive. Percolation also delivers power laws for the tortuosity that should be applicable whenever its expressions for flow or diffusion are verified. In these cases, however, the power is negative and the tortuosity diverges at the percolation threshold. If the purpose were just to find the limiting behavior of such functions for systems with large (or divergent) tortuosity, we would propose that the framework should be percolation theory. We have found, however, that some researchers find the limit of porosity equal to 1 more interesting than the opposite, small porosity limit. Further, many of the numerical studies are confined to large values of the porosity or small system sizes. In either of these cases, tortuosity values never grow very large, and the concept of a divergent porosity might appear to be extraneous. It should be possible, however, to apply concepts of finite-size scaling to address the limitations of the relatively small systems treated in models and simulations.

The remaining difficulty in proposing percolation theory as a unifying concept will probably be that so much of the research involves detailed comparisons of values of the tortuosity in systems far from the percolation threshold (near $\phi = 1$), where asymptotic formulations from percolation theory are not likely to be accurate. If the community evolves to attach less importance to the porosity limit of 1, which is not particularly applicable to

natural porous media, it is likely that we will indeed be able to apply the framework of percolation theory to develop a conceptual understanding of tortuosity.

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